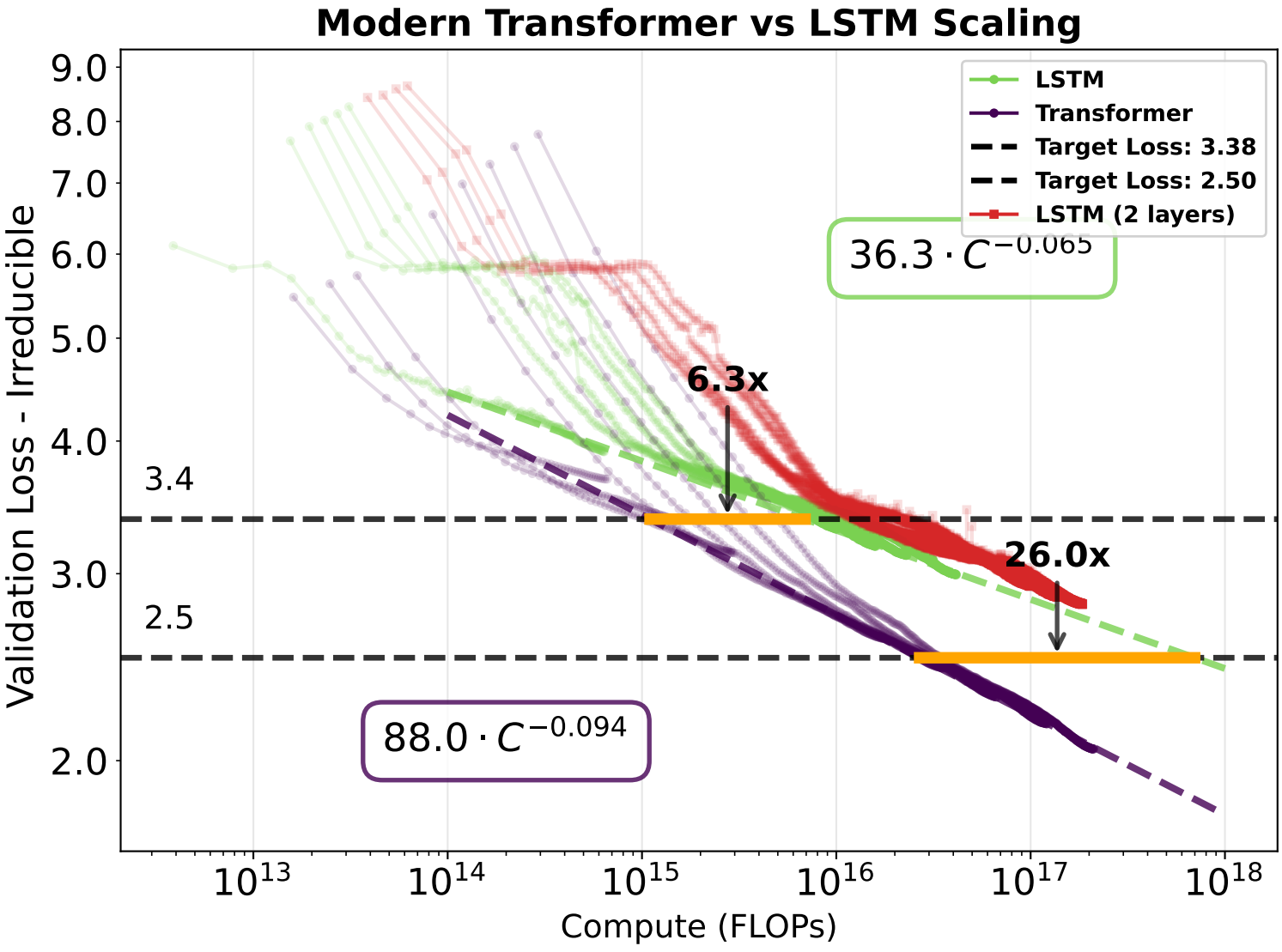
Modern Transformer vs LSTM Scaling
LSTM
Transformer
Target Loss: 3.38
Target Loss: 2.50
LSTM (2 layers)
$36.3 \cdot C^{-0.065}$
$88.0 \cdot C^{-0.094}$
6.3x
26.0x
3.4
2.5
Validation Loss - Irreducible
Compute (FLOPs)
9.0
8.0
7.0
6.0
5.0
4.0
3.0
2.0
10^{13}
10^{14}
10^{15}
10^{16}
10^{17}
10^{18}